Quadratic equations and parabolas

1

a
- i Maximum ii Negative iii 2 solutions

b
- i Minimum ii Positive iii 0 solutions

c
- i Maximum ii Negative iii 1 solution

d
- i Minimum ii Positive iii 2 solutions

2 The equation has no real solution.

3

a 2 solutions b 1 solution c 0 solutions

The discriminant and equations

4

a
- i $\Delta = 0$ ii 1

b
- i $\Delta = -76$ ii 0

c
- i $\Delta = 1$ ii 2

d
- i $\Delta = 16$ ii 2

e
- i $\Delta = 0$ ii 1

f
- i $\Delta = -23$ ii 0

g
- i $\Delta = 40$ ii 2

h
- i $\Delta = 100$ ii 2

5

a $\Delta = 0$ b Rational

6

a
- i $\Delta = -144$ ii Non-real roots.

b
- i $\Delta = 112$ ii Real, unequal and irrational roots.

c
- i $\Delta = 0$ ii Real, equal and rational roots.

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d

i $\Delta = 64$

ii Real, unequal and rational roots.

e

i $\Delta = 0$

ii Real, equal and rational roots.

f

i $\Delta = 65$

ii Real, unequal and irrational roots.

g

i $\Delta = -4$

ii Non-real roots.

h

i $\Delta = 3.75$

ii Real, unequal and irrational roots.

7

a $\Delta = 256$

b The roots are unequal and rational.

c $x = 12, x = -4$

8 There is only one real solution.

9 $b^2 - 4ac > 0$

10

a $9 + 8m$

b $16n^2 + 8$

c $45 - 4p$

d $16 + 4p^2$

e $12m$

f $\dfrac{-23n^2}{4}$

11 $a < \dfrac{9}{10}$

12 $m > -1$

13

a $m > -\dfrac{9}{20}$

b $m = 0$ must be eliminated as m is the coefficient of x^2 and by definition a quadratic equation must have an x^2 term.

14 $n = \pm 9$

15 $k = 1$

16 $m = -2, m = 4$

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17 **a**

i $64-8k$ ii $k=8$ iii $k \leq 8$ iv $k>8$

v $k<8$

b

i $400-8k$ ii $k=50$ iii $k \leq 50$ iv $k>50$

v $k<50$

c

i $52-12k$ ii $k=\frac{13}{3}$ iii $k \leq \frac{13}{3}$ iv $k>\frac{13}{3}$

v $k<\frac{13}{3}$

18

a $k>74$ **b** $k=75$

The discriminant and parabolas

19 None

20 One x-intercept

21

a i $a>0$ ii $\Delta>0$

b i $a<0$ ii $\Delta=0$

c i $a<0$ ii $\Delta<0$

d i $a>0$ ii $\Delta<0$

e i $a>0$ ii $\Delta<0$

f i $a>0$ ii $\Delta>0$

g i $a<0$ ii $\Delta=0$

h i $a<0$ ii $\Delta>0$

i i $a>0$ ii $\Delta=0$

j i $a<0$ ii $\Delta<0$

k i $a<0$ ii $\Delta=0$

l i $a<0$ ii $\Delta>0$

m i $a<0$ ii $\Delta<0$

n i $a>0$ ii $\Delta=0$

o i $a<0$ ii $\Delta>0$

p i $a>0$ ii $\Delta<0$

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22 a $m < -\frac{27}{4}$ b $m = -7$

23 $k = 16$

Application

24

a $0 < p < q$

b $x^2 + 2(p - q)x + p^2 - q^2 = 0$

c $\Delta = 8q(q - p)$

d Two real solutions as $q > p$, therefore the discriminant is positive.

e $x = 3p$

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